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Software Requirements Specification (SRS)

LexiFlow:

**A Web Platform for Translating and Transcribing
Native Mandarin Content**

Version 1.0

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Revision Page

a. Overview

This version of the document includes the requirement specification of the Lexiflow application covering the functional and non-functional requirements of the system.

b. Target Audience

- o Mandarin Language Learners (Students at UTM and other institutions)
- o Language Educators (UTM Language Academy lecturers and instructors)

c. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Adam Abouraya	First Version	26/06/2025

Note:

This Software Requirements Specification (SRS) template is adapted from IEEE Recommended Practice for Software Requirements Specification (SRS) (IEEE Std. 830-1998) that is simplified and customized to meet the need of SECJ3203 FYP1 SE course at Faculty of Computing, UTM. Examples of models are from Arlow and Neustadt (2002) and other sources stated accordingly. **This template is tailored for SRS in a plan-driven software development approach.**

Table of Contents

1	Introduction	4
1.1	Purpose	
1.2	Scope	
1.3	Definitions, Acronyms and Abbreviations	
1.4	References	
1.5	Overview	
2	Specific Requirements	
2.1	User characteristics	
2.2	System Features	
2.3	Use Case Details	
2.3.1	UC001: Use Case <Name of Use Case 1>	
	Use Case Specification of UC001	
	Activity Diagram of UC001	
	Sequence Diagram of UC001	
2.3.2	UC002: Use Case <Name of Use Case 2>	
	Use Case Specification of UC002	
	Activity Diagram of UC002	
	Sequence Diagram of UC002	
2.3. <i>n</i>	UC <i>n</i> : Use Case <Name of Use Case <i>n</i> >	
	Use Case Specification of UC00 <i>n</i>	
	Activity Diagram of UC00 <i>n</i>	
	Sequence Diagram of UC00 <i>n</i>	
2.4	Performance and Other Requirements	
2.5	Design Constraints	
3	Appendices	?

Appendix A: <Evidence of Requirements Elicitation Artefacts>

1. Introduction

1.1 Purpose

This SRS describes the functional and non-functional requirements for UniRide. The purpose of this document is to provide the requirements necessary to develop an end product that will meet the user requirements. These include the detailed descriptions of how a user will interact with the system and the system capabilities, which include integration points to functionalities provided by external services such as Google Maps and Firebase for real-time operations and data handling. This document proves to be the baseline reference for the development life-cycle stages, ensuring a smooth development process and in sync with the project's goals.

1.2 Scope

The proposed platform is designed to process long-form audio or video content, accepting inputs up to 120 minutes from user-uploaded files or YouTube URLs. It will employ voice activity detection (VAD) and end-of-sentence detection to segment the audio into linguistically meaningful, sentence-level chunks. For transcription, the platform will utilize automatic speech recognition (ASR), and translations will be generated via a Neural Machine Transformer (NMT) model. The output will be synchronized, timestamped bilingual subtitles, featuring original language text, target-language translations, and pinyin annotations, all produced by the transformer model. To facilitate efficient reuse, all processed subtitles will be stored in a searchable database, indexed by content identifiers like URLs, eliminating the need for reprocessing. Users will interact with the platform through a web interface, which allows for content uploads and customization of subtitle outputs.

1.3 Definitions, Acronyms and Abbreviation

Definitions of all terms, acronyms and abbreviation used are to be defined here.

- UTM – Universiti Teknologi Malaysia
- UI – User Interface
- API – Application Programming Interface
- SRS – Software Requirement Specification
- ASR – Automatic Speech Recognition

- NMT – Neural Machine Transformer
- VAD – Voice Activity Detection

1.4 References

1.5 Overview

The Software Requirements Specification (SRS) for UniRide has two main sections, which help to provide a general framework for implementing Lexiflow. The first section of this document is an Introduction that summarises the purpose and scope of the SRS and provides information about the high-level functionality of the software product.

Moving on to Section 2 which is the Specific Requirements, it forms the core of the document, and in this section it describes the functional requirements, system features, performance requirements, design constraints, software system attributes and properties. The section specifies every detail necessary for the development team during the design, development and testing of the Lexiflow application

2. Specific Requirements

2.1 User characteristics

For LexiFlow, there are two primary types of users: Mandarin Language Learners and Educators and Content Creators.

2.1.1 Mandarin Language Learners

Mandarin Language Learners are the core users of the LexiFlow platform, typically individuals who are proficient in English but seek to enhance their Mandarin skills, particularly in areas like tonal pronunciation and complex characters. They are expected to possess fundamental computer skills, including comfort with web-based applications, file uploads, and basic navigation within online interfaces. Their main objective is to efficiently improve their listening comprehension, overall understanding, and vocabulary through exposure to authentic Mandarin content. Their primary concern is accessing effective and engaging tools that support their language acquisition journey.

2.1.2 Educators and Content Creators

Educators and Content Creators represent a secondary user group who can utilize LexiFlow to develop more accessible and effective Mandarin learning materials. While their technical expertise may vary, they are expected to be capable of integrating the platform's processed content into their teaching methodologies or content creation workflows. Their focus is on facilitating language learning for others and producing high-quality, authentic language resources.

2.2 System Features

The LexiFlow platform is a standalone web-based software system designed to significantly enhance Mandarin language acquisition for learners. It operates as a comprehensive solution for processing long-form audio and video content, leveraging advanced technologies such as Automatic Speech Recognition (ASR), Neural Machine Translation (NMT), and Voice Activity Detection (VAD) to generate synchronized, bilingual subtitles with pinyin annotations. The system provides a user-friendly web interface for content submission and output customization, aiming to

simplify the process of engaging with authentic Mandarin materials and storing processed subtitles in a searchable database to facilitate instant reuse and continuous learning.

The system features is illustrated in Figure 2.1 below. The detail description of each module and functions is tabulated in Table 2.1.

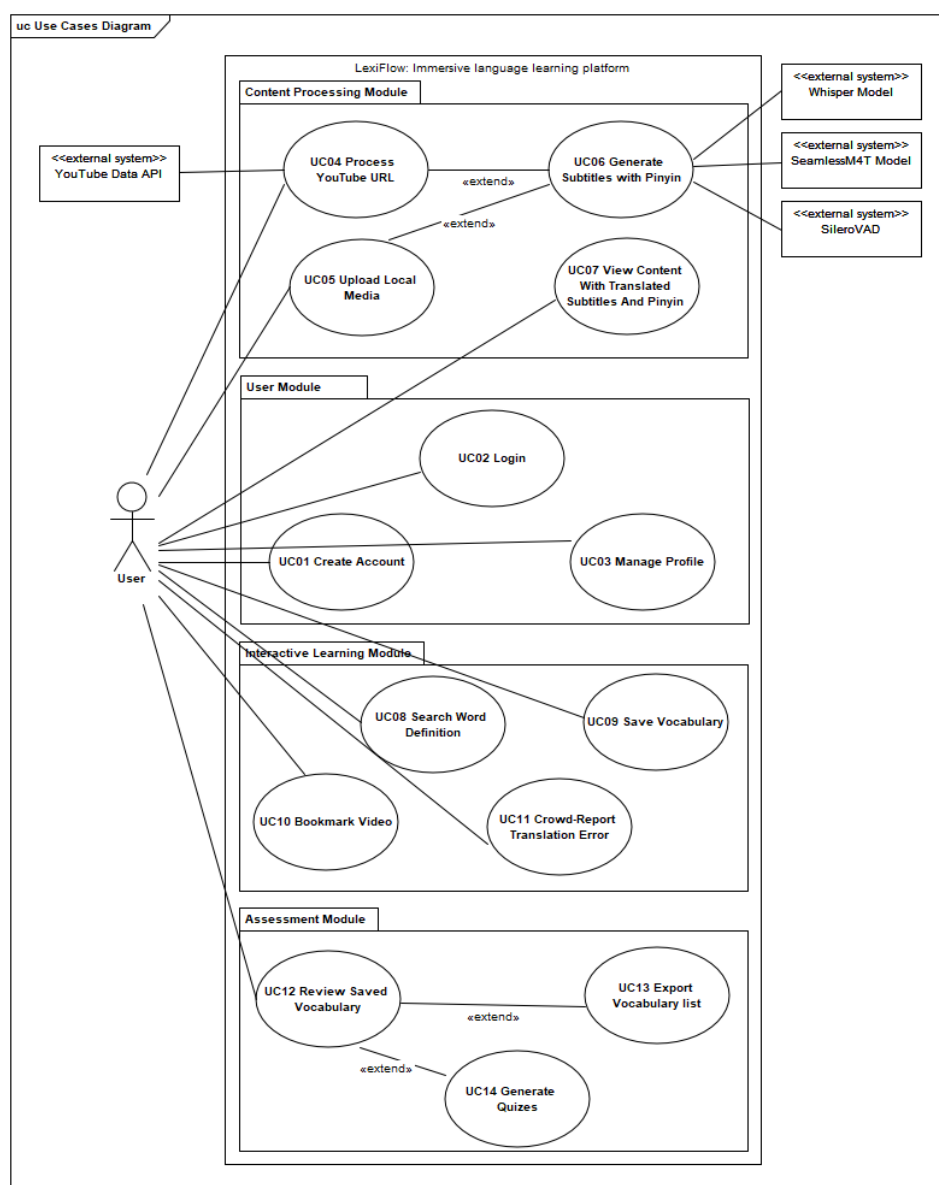


Figure 2.1: Use Case Diagram for LexiFlow

Table 2.1: Description of Module and Functions for LexiFlow

Module	Function	Description
External System	YouTube Data API	Integrates with YouTube for content processing.
	SileroVAD	Integrates with SileroVAD for content processing.

	Whisper	Integrates with Whisper for content processing.
	SeamlessM4Tv2	Integrates with SeamlessM4Tv2 for content processing.
User	UCO1 - Create Account	Allows students to register as users.
	UCO2 - Login	Authenticates users to access the platform.
	UCO3 - Manage Profile	Lets users update personal details/preferences.
Content Processing	UCO4 - Process YouTube URL	Fetches and processes video content from a YouTube URL.
	UCO6 - Generate Subtitles	Automatically generates subtitles for video content.
	UCO5 - Upload Local Media	Allows users to upload their own media files for processing.
	UCO7 - Find Translated Content	Searches for videos with translated subtitles.
Interactive learning	UCO8 - Search Word Definition	Provides definitions of words from subtitles/videos.
	UCO9 - Save Vocabulary	Saves selected words/phrases to a personal list.
	UC10 - Bookmark Video	Marks videos for later review.
	UC11 - Crowd Report Translation Error	Users can report subtitle translation errors for community review.
Assessment	UC12 - Review Saved Vocabulary	Displays previously saved words/phrases for revision.
	UCI3 - Export Vocabulary List	Exports vocabulary lists to external files (e.g., PDF, CSV).
	UCI4 - Generate Quizzes	Creates quizzes based on saved vocabulary.

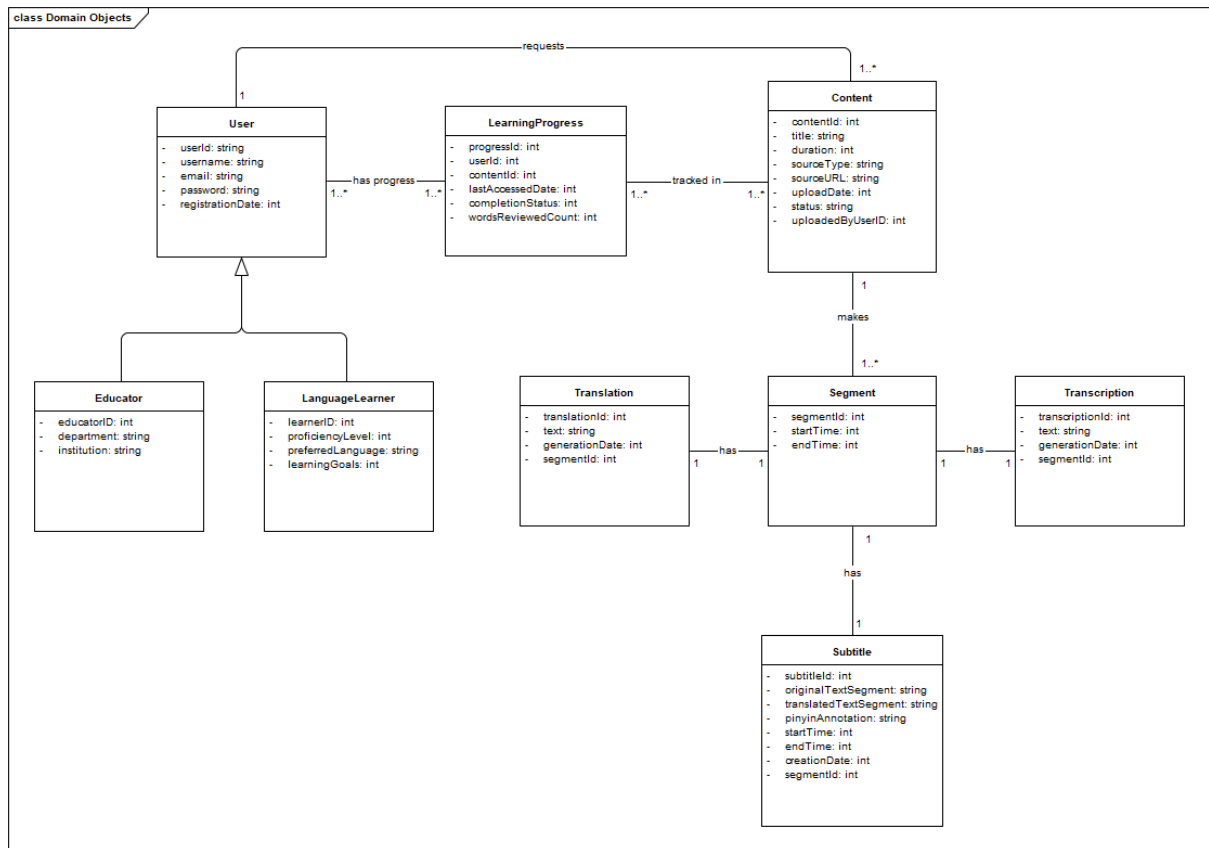


Figure 2.2: Domain Model for LexiFlow

2.3 Use Case Details

2.3.1 UC001: Use Case <Create Account>

Table 2.2: Use Case Description for <Create Account>

Use Case ID	UC01
Use Case Name	Create Account
Description	This use case outlines the process for a prospective user to create an account within the LexiFlow platform, allowing them to access personalized features and utilize the system's functionalities.
Actor(s)	User (Prospective Language Learner or Educator)
Pre-condition(s)	1. The user does not have an existing account in the LexiFlow system. 2. The user has access to the LexiFlow web interface.
Normal Flow(s)- NF	1. The user navigates to the LexiFlow registration page. 2. The system displays the account registration form. 3. The user enters all required registration details, including full name, email address, desired password, confirms the password, and selects

	their user type (e.g., Language Learner or Educator). 4. The system performs validation checks on the entered data (e.g., email format, password strength, matching passwords). 5. The system creates a new user account with the provided details and stores it in the database. 6. The system displays a success message and optionally sends a confirmation/verification email to the registered email address. 7. The system redirects the user to the login page or directly logs them into their new account.
Alternative Flow(s) - AF	AF1. Existing Email: 4a. If the system detects that the entered email address is already registered: 4a1. The system displays an error message indicating that the email is already in use and suggests logging in or using a different email. 4a2. The use case ends, prompting the user to either log in or restart the registration with a new email
Exception Flow(s) - EF	EF1. Database Write Error: 5a. If the system encounters an error while attempting to save the new user account to the database: 5a1. The system logs the technical error for administrative review. 5a2. The system displays a generic error message to the user, instructing them to try again later. 5a3. The use case terminates unsuccessfully. EF2. Network Interruption: 3a. If a network connectivity issue occurs during the user's submission of the form: 3a1. The system's request times out or indicates a network error. 3a2. The user is prompted to check their internet connection and retry the submission. 3a3. The use case may restart from step 3 or terminate, depending on the severity of the interruption.
Post-condition(s)	1. A new User account record is successfully created and stored in the LexiFlow database. 2. The user is either logged in or presented with an option to log in to their newly created account. 3. (Optional) A confirmation/verification email is sent to the user's registered email address.

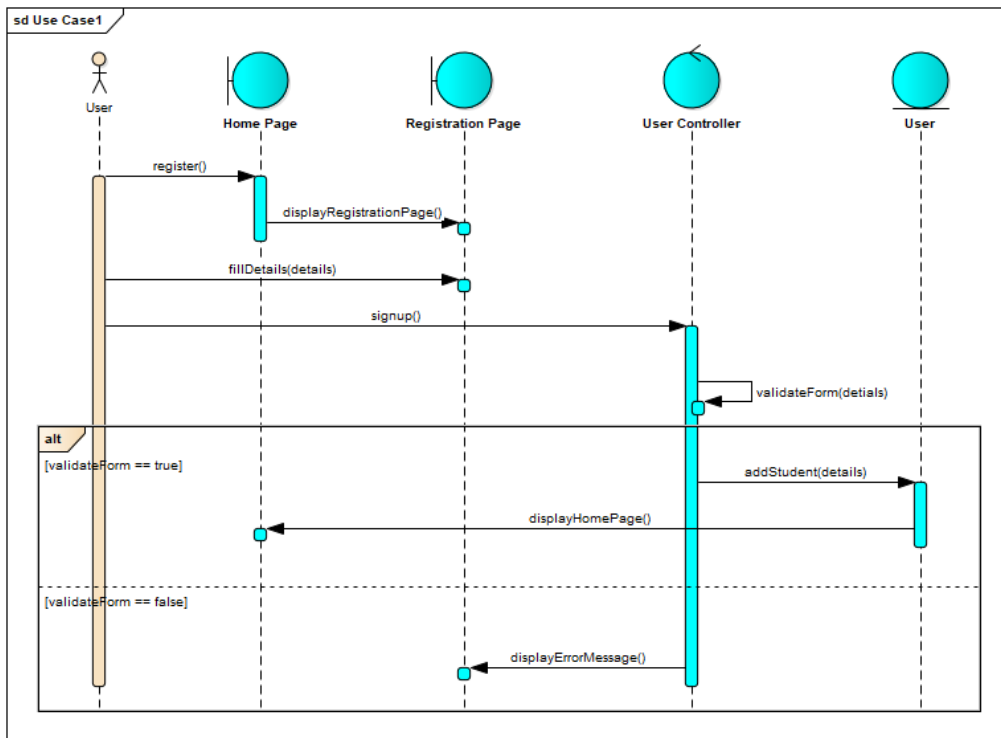


Figure 3.1: Sequence Diagram for <Create Account>

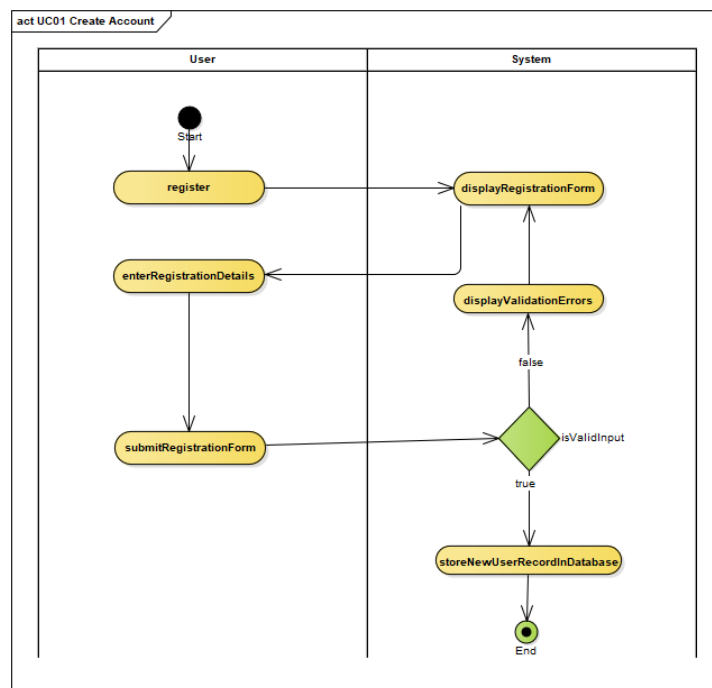


Figure 3.2: Activity Diagram for < Create Account >

2.3.2 UC002: Use Case <Login>

Use Case ID	UC02
Use Case Name	Login

Description	This use case enables a registered user to securely log into their LexiFlow account, gaining access to personalized content and system features.
Actor(s)	User (Registered Language Learner or Educator)
Pre-condition(s)	1. The user has an existing and active account within the LexiFlow system. 2. The user has access to the LexiFlow web interface.
Normal Flow(s)- NF	1. The user navigates to the LexiFlow login page. 2. The system displays the login form requesting email/username and password. 3. The user enters their registered email/username and password. 4. The system validates the entered credentials against the stored user data. 5. Upon successful validation, the system creates a user session. 6. The system redirects the user to their personalized dashboard or the last accessed page.
Alternative Flow(s) - AF	AF1. Forgot Password: 3a. The user clicks on the "Forgot Password" link. 3a1. The system prompts the user to enter their registered email address. 3a2. The user enters their email. 3a3. The system sends a password reset link or code to the registered email address. 3a4. The use case ends, requiring the user to follow the email instructions.
Exception Flow(s) - EF	EF1. Invalid Credentials: 4a. If the entered email/username and password do not match a registered account or are incorrect: 4a1. The system displays an error message indicating "Invalid credentials" or "Incorrect email or password." 4a2. The system prompts the user to re-enter their credentials or use the "Forgot Password" option. 4a3. The use case ends, requiring the user to retry. EF2. Account Locked/Suspended: 4b. If the account is locked or suspended due to multiple failed login attempts or administrative action: 4b1. The system displays an error message indicating the account status and provides instructions for resolution (e.g., contact support, wait for lockout period). 4b2. The use case terminates unsuccessfully.
Post-condition(s)	1. The user is successfully authenticated and logged into their LexiFlow account. 2. A user session is established. 3. The user has access to their personalized content and system functionalities.

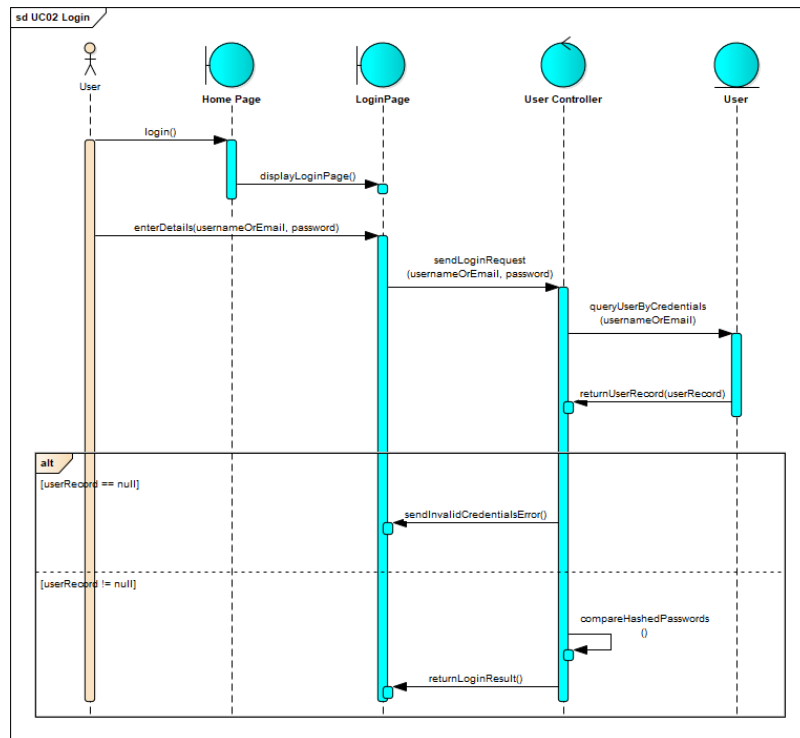


Figure 3.3: Sequence Diagram for <Login>

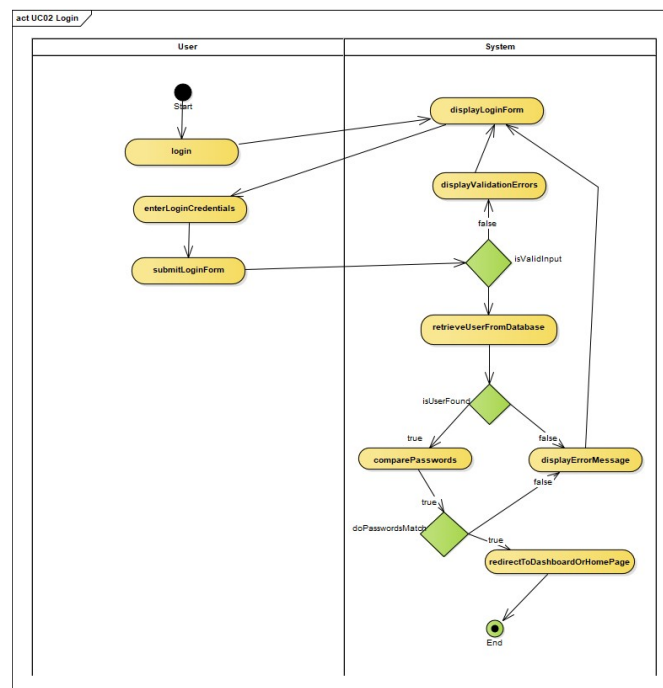


Figure 3.4: Activity Diagram for < Login >

2.3.3 UC3: Use Case <Manage Profile>

Use Case ID	UC03
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Use Case Name	Manage Profile
Description	This use case allows an authenticated user to view, update, and manage their personal information and account settings within the LexiFlow system.
Actor(s)	User (Authenticated Language Learner or Educator)
Pre-condition(s)	1. The user must be successfully logged into their LexiFlow account.
Normal Flow(s)- NF	1. The user navigates to the "Manage Profile" or "Settings" section of the LexiFlow application. 2. The system retrieves and displays the user's current profile information (e.g., name, email address, password fields, user type). 3. The user modifies the desired profile fields (e.g., updates their name, changes their password, or updates their user type preference). 4. The user submits the updated information. 5. The system performs validation checks on the submitted data (e.g., ensuring email format is correct, new password meets strength criteria and matches confirmation). 6. The system updates the user's profile record in the database with the new information. 7. The system displays a confirmation message indicating that the profile has been successfully updated.
Alternative Flow(s) - AF	AF1. Change Password: 3a. If the user only wishes to change their password: 3a1. The user initiates the password change process within the profile section. 3a2. The system prompts the user to enter their current password, a new password, and to confirm the new password. 3a3. The user enters the required password details. 3a4. The system validates the current password and ensures the new password meets security criteria and matches its confirmation. 3a5. The system updates the user's password in the database. 3a6. The system displays a success message for the password change.
Exception Flow(s) - EF	EF1. Invalid Input: 5a. If any of the updated information (e.g., email, new password) fails validation checks: 5a1. The system highlights the invalid fields and displays specific error messages (e.g., "Invalid email format," "Password too weak," "Passwords do not match"). 5a2. The user is prompted to correct the errors. 5a3. The use case continues from step 3. EF2. Database Update Error: 6a. If the system encounters a technical error while attempting to save the updated profile to the database:

	6a1. The system logs the error for administrative review. 6a2. The system displays a generic error message to the user, suggesting they try again later. 6a3. The use case terminates unsuccessfully, and the profile remains unchanged.
Post-condition(s)	1. The user's profile information (excluding password, if only other fields were changed) is successfully updated in the LexiFlow system. 2. If the password was changed, the new password is now active for the user's account.

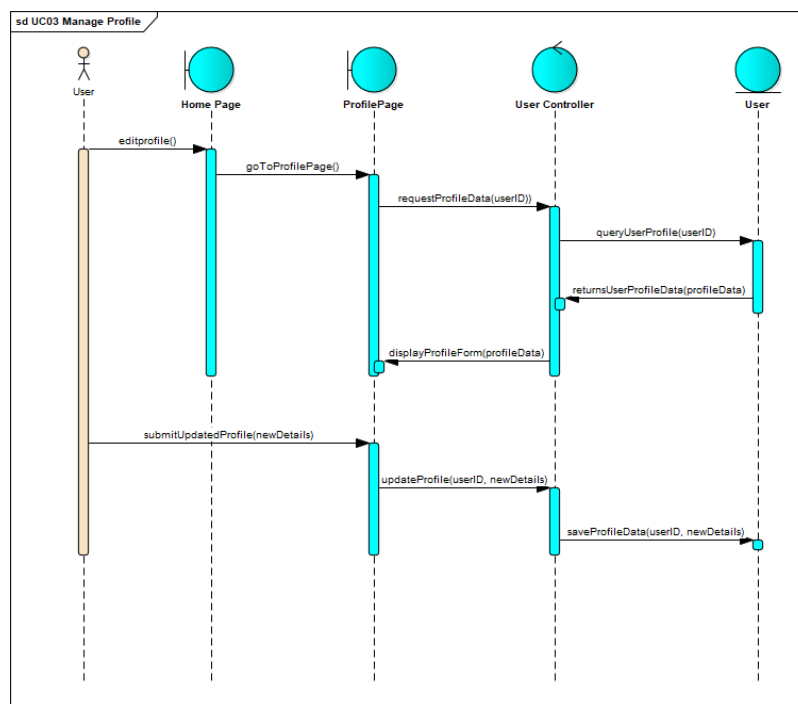


Figure 3.5: Sequence Diagram for <Manage Profile>

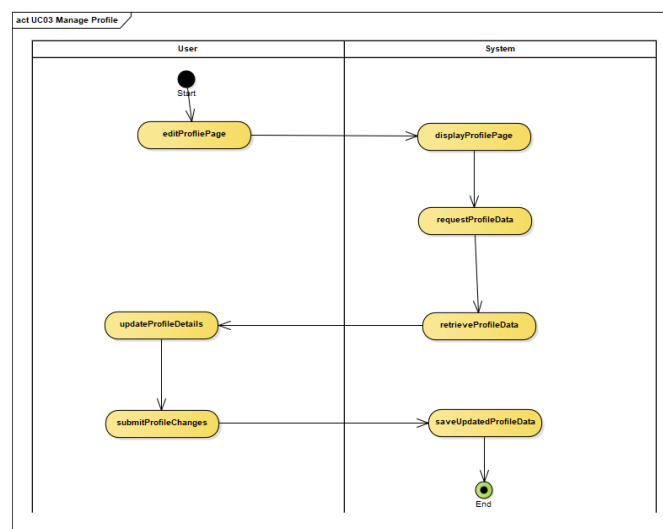


Figure 3.6: Activity Diagram for < Manage Profile>

2.3.4 UC4: Use Case <Process YouTube URL>

Use Case ID	UC04
Use Case Name	Process YouTube URL
Description	This use case allows an authenticated user to submit a YouTube video URL for automated transcription, translation, and synchronized subtitle generation within LexiFlow.
Actor(s)	User (Authenticated Language Learner or Educator), YouTube Data API
Pre-condition(s)	1. The user is successfully logged into their LexiFlow account. 2. The LexiFlow system has internet access to reach YouTube.
Normal Flow(s)- NF	1. The user navigates to the "Upload/Process Content" section. 2. The user selects the option to process a YouTube URL. 3. The user inputs a valid YouTube video URL into the provided field. 4. The user submits the URL for processing. 5. The system validates the URL and attempts to fetch the video's audio/video stream. 6. The system then initiates the content processing pipeline. 7. The system notifies the user that processing has started and will notify them upon completion.
Alternative Flow(s) - AF	AF1. Invalid YouTube URL: 5a. If the entered URL is not a valid YouTube URL or points to a non-existent video: 5a1. The system displays an error message indicating "Invalid YouTube URL" or "Video not found." 5a2. The user is prompted to enter a correct URL. 5a3. The use case continues from step 3. AF2. Regionally Restricted/Private Video: 5b. If the video is geographically restricted or set to private and cannot be accessed by the system: 5b1. The system displays an error message stating the video cannot be processed due to access restrictions. 5b2. The use case terminates unsuccessfully.
Exception Flow(s) - EF	EF1. Processing Pipeline Failure: 6a. If any stage of the content processing pipeline (e.g., ASR, NMT, synchronization) encounters a critical error: 6a1. The system logs the detailed error for technical review. 6a2. The system displays a generic error message to the user, indicating that processing failed and asking them to try again later or contact support. 6a3. The use case terminates unsuccessfully, and no subtitles are generated. EF2. Network Error During Fetch:

	5c. If a network connectivity issue prevents the system from fetching the YouTube video: 5c1. The system displays a network error message. 5c2. The user is prompted to check their internet connection or retry later. 5c3. The use case terminates unsuccessfully.
Post-condition(s)	1. The YouTube video's audio/video content is successfully extracted by LexiFlow. 2. The content processing pipeline is initiated for transcription, translation, and subtitle generation. 3. The user receives a notification regarding the processing status.

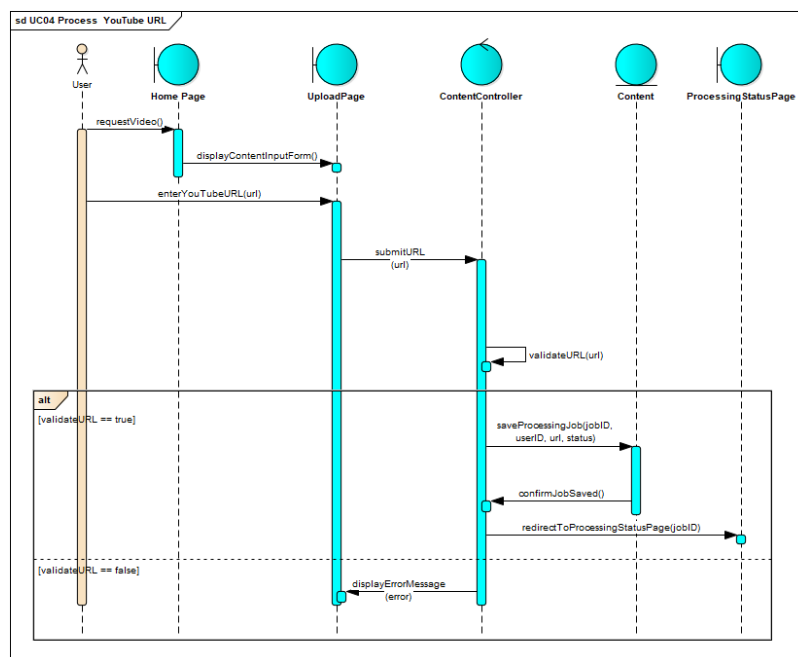


Figure 3.7: Sequence Diagram for <Process YouTube URL>

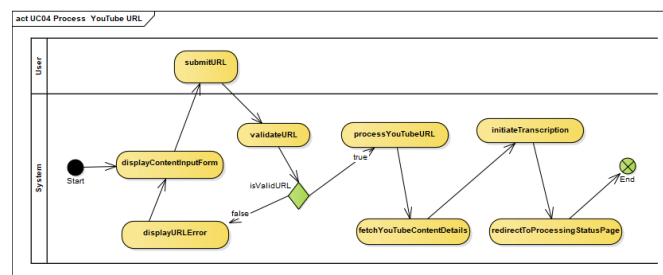


Figure 3.8: Activity Diagram for < Process YouTube URL>

2.3.5 UC5: Use Case <Upload Local Media>

Use Case ID	UC05
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Use Case Name	Upload Local Media
Description	This use case allows an authenticated user to upload a local audio or video file from their device to the LexiFlow system for processing, transcription, translation, and subtitle generation.
Actor(s)	User (Authenticated Language Learner or Educator)
Pre-condition(s)	1. The user is successfully logged into their LexiFlow account. 2. The user has a local audio or video file available on their device.
Normal Flow(s)- NF	1. The user navigates to the "Upload Content" section. 2. The user selects the option to upload a local media file. 3. The system displays a file upload interface (e.g., a file browser dialog). 4. The user selects the desired audio or video file from their local device. 5. The user initiates the upload process. 6. The system uploads the file to the LexiFlow server. 7. Upon successful upload, the system initiates the content processing pipeline. 8. The system notifies the user that processing has started and will notify them upon completion.
Alternative Flow(s) - AF	AF1. Unsupported File Format: 5a. If the selected file's format is not supported by LexiFlow: 5a1. The system displays an error message indicating "Unsupported file format." 5a2. The user is prompted to upload a file in a supported format. 5a3. The use case ends, requiring the user to select a different file. AF2. File Size Exceeds Limit: 5b. If the selected file's size exceeds the maximum allowed upload limit: 5b1. The system displays an error message indicating "File size limit exceeded." 5b2. The user is prompted to upload a smaller file or split the content. 5b3. The use case ends, requiring the user to select a different file.
Exception Flow(s) - EF	EF1. Upload Interruption/Failure: 6a. If the file upload process is interrupted (e.g., network disconnect) or fails: 6a1. The system displays an error message indicating "Upload failed" or "Network error." 6a2. The user is prompted to retry the upload. 6a3. The use case terminates unsuccessfully. EF2. Processing Pipeline Failure: 7a. If any stage of the content processing pipeline (e.g., ASR, NMT, synchronization) encounters a critical error after upload: 7a1. The system logs the detailed error for technical review. 7a2. The system displays a generic error message to the user,

	indicating that processing failed and asking them to try again later or contact support. 7a3. The use case terminates unsuccessfully, and no subtitles are generated.
Post-condition(s)	1. The local media file is successfully uploaded and stored on the LexiFlow server. 2. The content processing pipeline is initiated for transcription, translation, and subtitle generation. 3. The user receives a notification regarding the processing status.

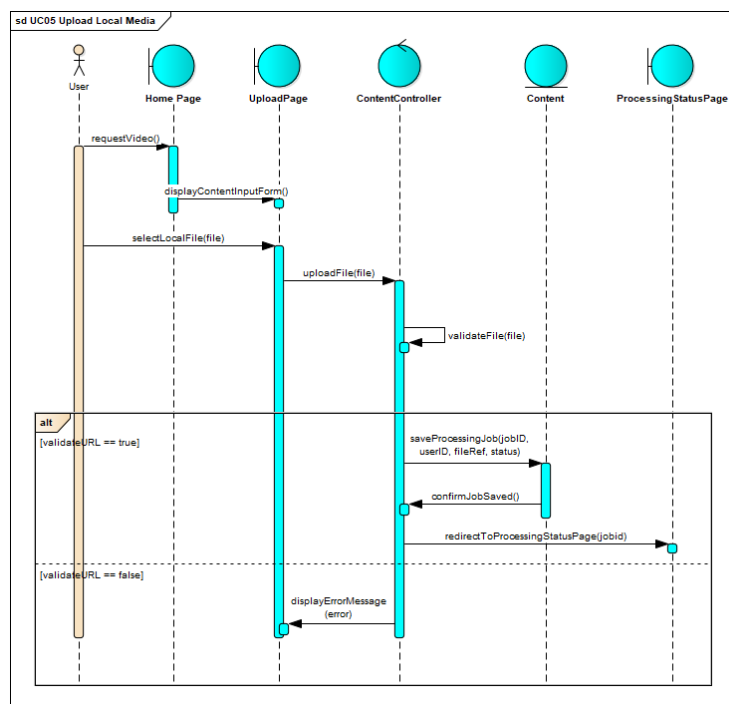


Figure 3.9: Sequence Diagram for <Upload Local Media>

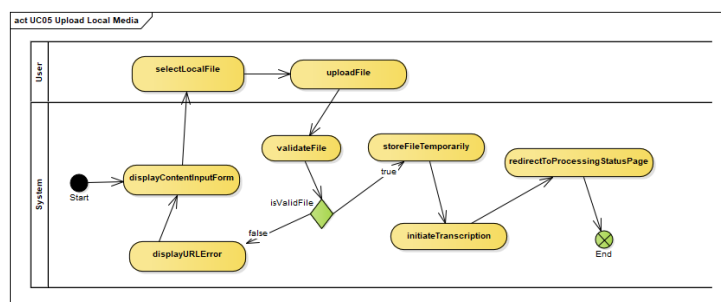


Figure 3.10: Activity Diagram for < Upload Local Media>

2.3.6 UC6: Use Case <Generate Subtitles>

Use Case ID	UC06
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Use Case Name	Generate Subtitles
Description	This use case describes the automated process by which the LexiFlow system generates accurate, timestamped, and bilingual subtitles (original and translated with pinyin) for processed audio/video content. This use case is typically triggered internally after a user's content upload (UC05) or YouTube URL submission (UC04).
Actor(s)	LexiFlow System (triggered by User interaction)
Pre-condition(s)	1. Audio/video content has been successfully uploaded or fetched by the LexiFlow system. 2. The content is available for processing. 3. Necessary AI/ML services (ASR, NMT) are available and operational.
Normal Flow(s)- NF	1. The LexiFlow system receives a request to process a piece of content. 2. The system performs Voice Activity Detection (VAD) on the audio stream to identify speech segments and silences. 3. The system sends the identified speech segments to the Automatic Speech Recognition (ASR) service to generate a transcription in the original language. 4. The system receives the original language transcription from the ASR service. 5. The system sends the original language transcription to the Neural Machine Translation (NMT) service for translation into the target language (English, and Pinyin for Mandarin characters). 6. The system receives the translated text and pinyin from the NMT service. 7. The system synchronizes the original transcription and translated text with the audio/video timestamps, creating segments for subtitles. 8. The system formats the synchronized text and pinyin into a subtitle file (e.g., SRT, VTT). 9. The system stores the generated subtitle file(s) and metadata in the database, linked to the original content. 10. The system notifies the user that the subtitles are ready.
Alternative Flow(s) - AF	AF1. No Speech Detected: 2a. If VAD detects no discernible speech in the content: 2a1. The system generates an empty subtitle file or a file indicating "No speech detected." 2a2. The use case proceeds to store this result and notify the user.
Exception Flow(s) - EF	EF1. ASR Service Failure: 3a. If the ASR service fails to transcribe the audio or returns an error: 3a1. The system logs the ASR error. 3a2. The system attempts a retry (if configured) or marks the content as partially processed/failed. 3a3. The use case terminates unsuccessfully for this stage, potentially notifying the user of partial failure. EF2. NMT Service Failure:

	5a. If the NMT service fails to translate the text or returns an error: 5a1. The system logs the NMT error. 5a2. The system attempts a retry or generates subtitles with only the original language. 5a3. The use case continues to synchronization, potentially with incomplete data.
Post-condition(s)	1. Timestamped original language subtitles are generated. 2. Timestamped translated subtitles (with pinyin for Mandarin) are generated. 3. Subtitle files and associated metadata are stored in the LexiFlow database. 4. The user is notified that the subtitles are available.

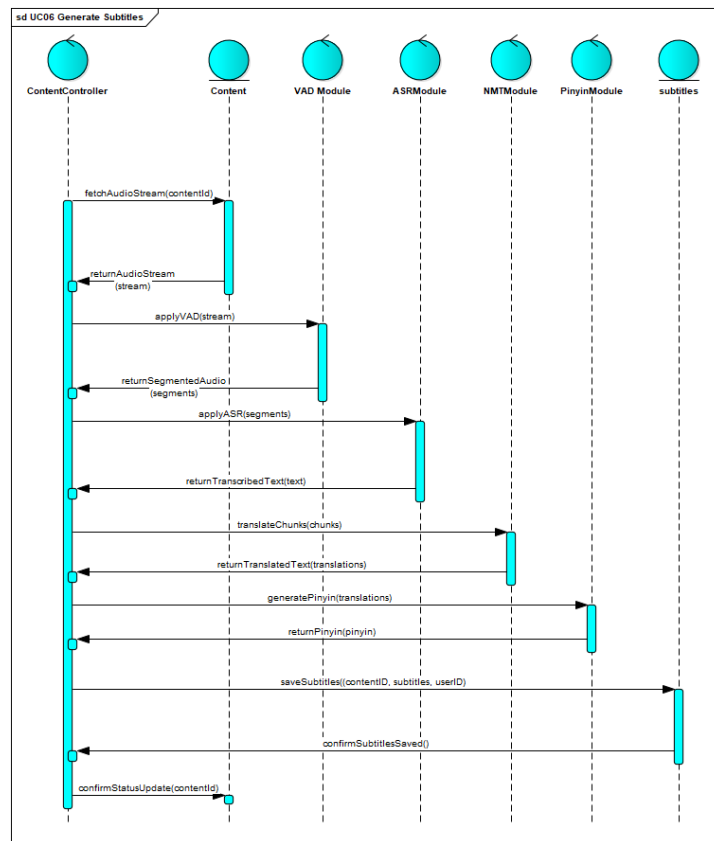


Figure 3.11: Sequence Diagram for <Generate Subtitles>

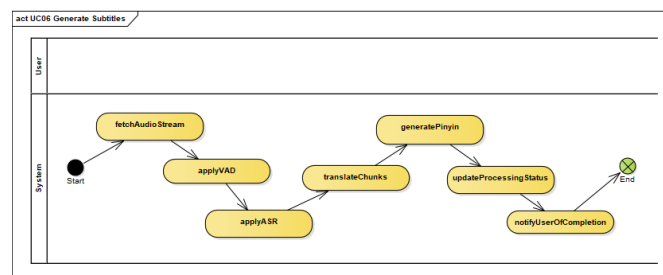


Figure 3.12: Activity Diagram for <Generate Subtitles>

2.3.7 UC7: Use Case <Find Translated Content>

Use Case ID	UC07
Use Case Name	Find Translated Content
Description	This use case enables an authenticated user to search for and retrieve previously processed audio/video content and its associated bilingual subtitles within the LexiFlow system.
Actor(s)	User (Authenticated Language Learner or Educator)
Pre-condition(s)	1. The user is successfully logged into their LexiFlow account. 2. There is content that has been previously processed and had subtitles generated and stored in the system.
Normal Flow(s)- NF	1. The user navigates to the "My Content" or "Search" section. 2. The user enters search keywords (e.g., video title, topic, specific vocabulary) into the search bar. 3. The user initiates the search. 4. The system queries the database for content matching the search criteria (e.g., in content titles, descriptions, or within the subtitle text). 5. The system retrieves and displays a list of search results, including relevant metadata (e.g., content title, duration, processing date). 6. The user selects a desired piece of content from the search results. 7. The system displays the selected content along with its synchronized bilingual subtitles.
Alternative Flow(s) - AF	AF1. No Search Results: 5a. If the search query yields no matching content: 5a1. The system displays a message indicating "No content found for your search" and may suggest refining the query. 5a2. The use case ends, prompting the user to try a different search.
Exception Flow(s) - EF	EF1. Database Query Error: 4a. If the system encounters a technical error during the database search query: 4a1. The system logs the error for administrative review. 4a2. The system displays a generic error message to the user, suggesting they try again later. 4a3. The use case terminates unsuccessfully.
Post-condition(s)	1. The user successfully finds and views the desired processed content and its associated bilingual subtitles. 2. The user can interact with the subtitles (e.g., click on words, save vocabulary).

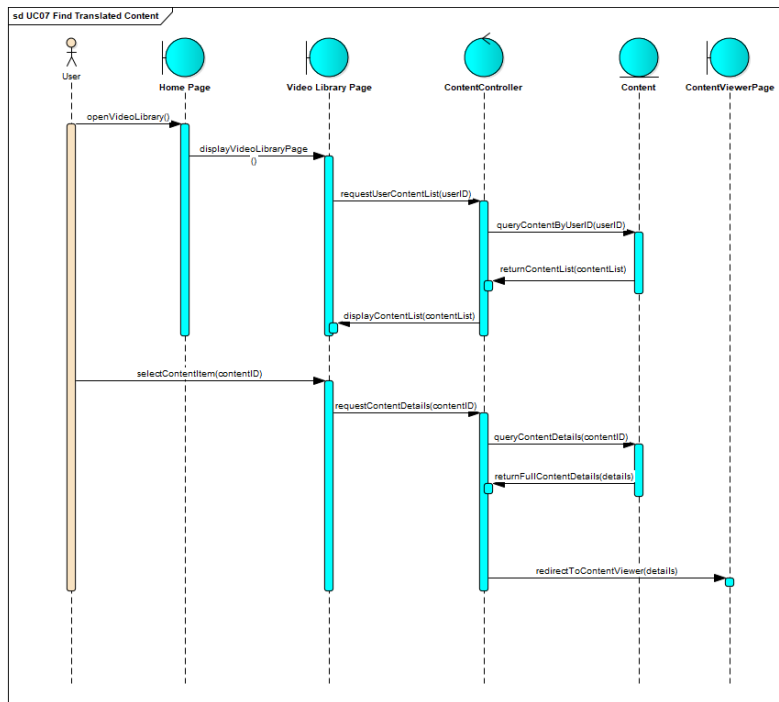


Figure 3.13: Sequence Diagram for <Find Translated Content>

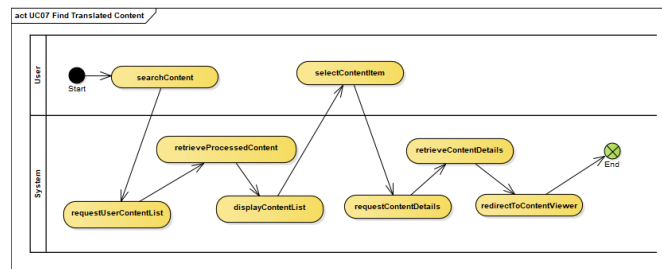


Figure 3.14: Activity Diagram for <Find Translated Content>

2.3.8 UC8: Use Case <Search Word Definition>

Use Case ID	UC08
Use Case Name	Search Word Definition
Description	This use case allows an authenticated user to quickly look up the definition of a specific word or character directly from the displayed subtitles, including its pinyin pronunciation.
Actor(s)	User (Authenticated Language Learner or Educator)
Pre-condition(s)	1. The user is successfully logged into their LexiFlow account. 2. The user is currently viewing content with synchronized subtitles.
Normal Flow(s)- NF	1. While viewing subtitles, the user hovers over or clicks on a specific word or character in the Mandarin (original or pinyin) part of the subtitle.

	2. The system identifies the selected word/character. 3. The system queries its internal dictionary or an integrated external dictionary service for the definition and pinyin. 4. The system displays the definition and pinyin (and potentially example sentences or usage notes) in a pop-up window or a dedicated side panel. 5. The user reviews the definition.
Alternative Flow(s) - AF	AF1. Word Not Found: 4a. If the system cannot find a definition for the selected word/character in its dictionary: 4a1. The system displays a message indicating "Definition not found" or "Word not in dictionary." 4a2. The use case ends, and no definition is displayed.
Exception Flow(s) - EF	EF1. Dictionary Service Error: 3a. If the internal or external dictionary service encounters an error during the lookup: 3a1. The system logs the error. 3a2. The system displays a temporary error message (e.g., "Could not retrieve definition at this time"). 3a3. The use case terminates unsuccessfully.
Post-condition(s)	1. The user successfully views the definition and pinyin of the selected word/character.

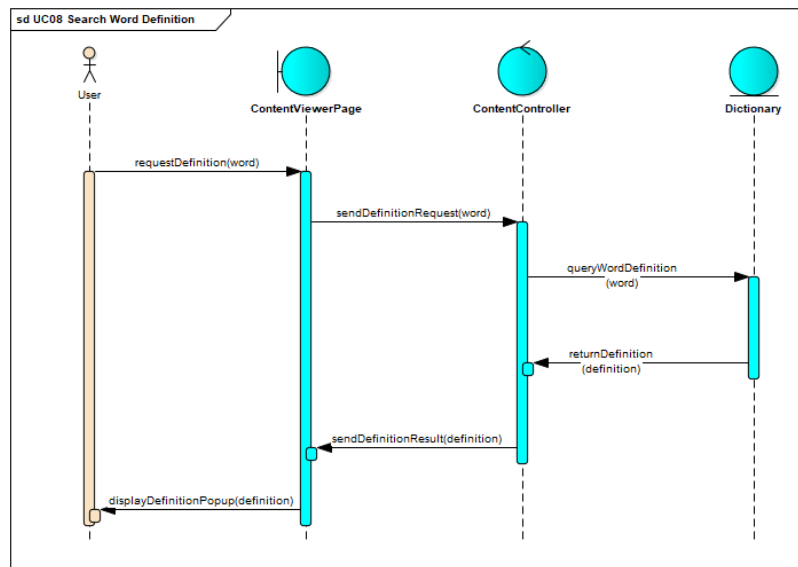


Figure 3.15: Sequence Diagram for <Search Word Definition>

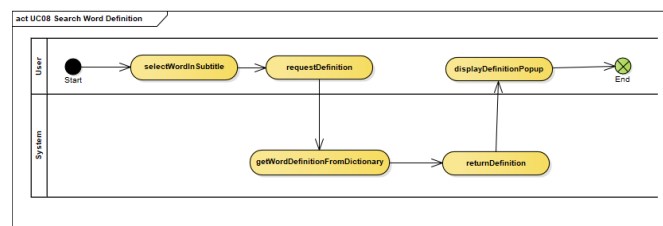


Figure 3.16: Activity Diagram for <Search Word Definition>

2.3.9 UC9: Use Case <Save Vocabulary>

Use Case ID	UC09
Use Case Name	Save Vocabulary
Description	This use case allows an authenticated user to save specific words or phrases from the subtitles they are viewing into a personal vocabulary list for later review and learning activities.
Actor(s)	User (Authenticated Language Learner or Educator)
Pre-condition(s)	1. The user is successfully logged into their LexiFlow account. 2. The user is currently viewing content with synchronized subtitles.
Normal Flow(s)- NF	1. While viewing subtitles, the user selects a word or phrase they wish to save. 2. The user clicks a "Save Vocabulary" or equivalent button/option associated with the selected text. 3. The system identifies the selected word/phrase along with its definition, pinyin, and context from the subtitle. 4. The system adds the word/phrase and its associated details to the user's personal vocabulary list in the database. 5. The system provides visual feedback (e.g., a "Saved!" message, highlight change) to confirm the addition.
Alternative Flow(s) - AF	AF1. Word Already Saved: 4a. If the selected word/phrase is already present in the user's vocabulary list: 4a1. The system displays a message indicating "Already saved" or "Item already in your vocabulary." 4a2. The use case ends, no duplicate entry is created.
Exception Flow(s) - EF	EF1. Database Write Error: 4b. If the system encounters a technical error while attempting to save the vocabulary item to the database: 4b1. The system logs the error. 4b2. The system displays a generic error message (e.g., "Could not save vocabulary. Please try again."). 4b3. The use case terminates unsuccessfully.
Post-condition(s)	1. The selected word or phrase is successfully added to the user's personal vocabulary list. 2. The vocabulary item is available for future review and learning activities.

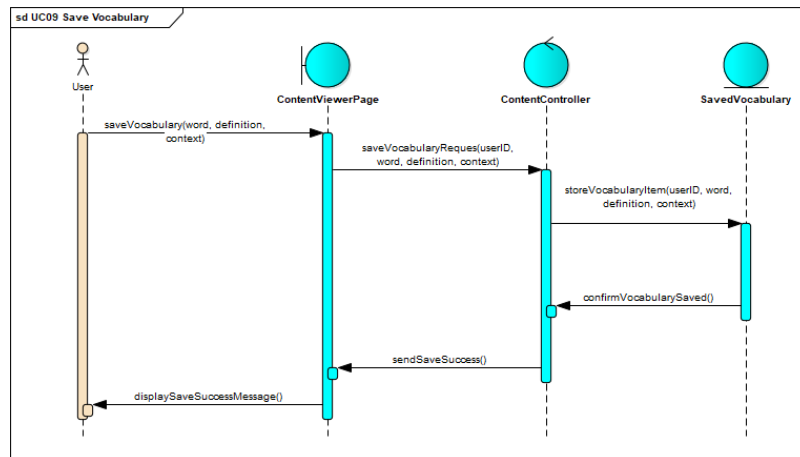


Figure 3.17: Sequence Diagram for <Save Vocabulary>

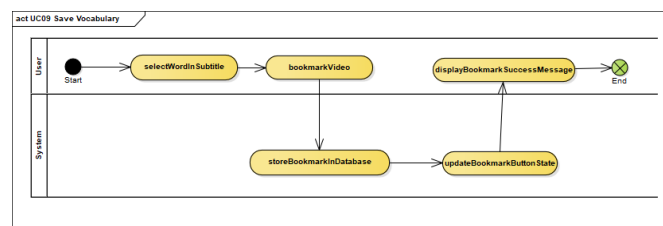


Figure 3.18: Activity Diagram for <Save Vocabulary>

2.3.10 UC10: Use Case <Bookmark Video>

Use Case ID	UC10
Use Case Name	Bookmark Video
Description	This use case allows an authenticated user to bookmark specific videos or sections of videos for quick and easy access later, facilitating review and focused study.
Actor(s)	User (Authenticated Language Learner or Educator)
Pre-condition(s)	1. The user is successfully logged into their LexiFlow account. 2. The user is currently viewing content within the LexiFlow player.
Normal Flow(s)- NF	1. While viewing a video, the user clicks on a "Bookmark Video" or similar icon/option. 2. The system records the video ID and optionally the current timestamp. 3. The system adds this bookmark to the user's personal bookmark list in the database. 4. The system provides visual confirmation (e.g., "Video Bookmarked").
Alternative Flow(s) - AF	AF1. Video Already Bookmarked: 3a. If the video is already present in the user's bookmark list: 3a1. The system displays a message indicating "Already

	bookmarked" or "Video is already in your bookmarks." 3a2. The use case ends, no duplicate entry is created.
Exception Flow(s) - EF	EF1. Database Write Error: 3b. If the system encounters a technical error while attempting to save the bookmark to the database: 3b1. The system logs the error. 3b2. The system displays a generic error message (e.g., "Could not bookmark video. Please try again."). 3b3. The use case terminates unsuccessfully.
Post-condition(s)	1. The selected video (and optionally specific timestamp) is successfully added to the user's personal bookmark list. 2. The bookmarked video is accessible from the user's profile for quick access.

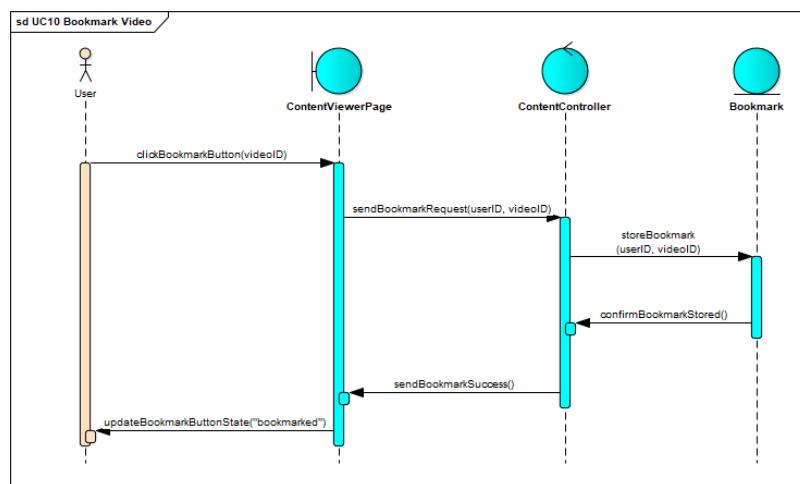


Figure 3.19: Sequence Diagram for <Bookmark Video>

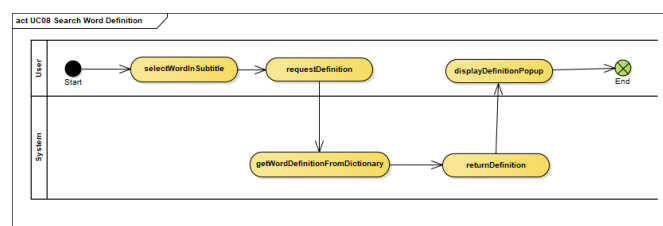


Figure 3.20: Activity Diagram for <Bookmark Video>

2.3.11 UC11: Use Case <Crowd Report Translation Error>

Use Case ID	UC11
Use Case Name	Crowd Report Translation Error

Description	This use case enables an authenticated user to report inaccuracies or suggest improvements for translated subtitles, contributing to the system's ongoing refinement through crowdsourced feedback.
Actor(s)	User (Authenticated Language Learner or Educator)
Pre-condition(s)	1. The user is successfully logged into their LexiFlow account. 2. The user is currently viewing content with synchronized, translated subtitles.
Normal Flow(s)- NF	1. While viewing translated subtitles, the user identifies a potential translation error or area for improvement. 2. The user selects the specific subtitle segment containing the error. 3. The user initiates the "Report Error" function (e.g., by right-clicking or clicking an icon). 4. The system presents a form where the user can describe the error and optionally suggest a corrected translation or provide comments. 5. The user submits the report. 6. The system captures the reported error, associated subtitle ID, timestamp, user ID, and suggested correction/comment. 7. The system stores the error report in a dedicated database for review by administrators or content moderators. 8. The system provides a confirmation message (e.g., "Report submitted. Thank you for your contribution!").
Alternative Flow(s) - AF	AF1. Report Already Exists: 6a. If a similar report for the exact subtitle segment has recently been submitted by the same user or other users: 6a1. The system may prompt the user if they wish to add to the existing report or indicate that it is already noted. 6a2. The use case may proceed to record the new input as an additional comment or terminate without creating a duplicate.
Exception Flow(s) - EF	EF1. Submission Failure: 6b. If the system encounters an error during the submission or storage of the report: 6b1. The system logs the error. 6b2. The system displays an error message to the user (e.g., "Could not submit report. Please try again later."). 6b3. The use case terminates unsuccessfully.
Post-condition(s)	1. A translation error report, including user feedback and context, is successfully recorded in the LexiFlow database. 2. The report is available for review by content administrators or moderation teams.

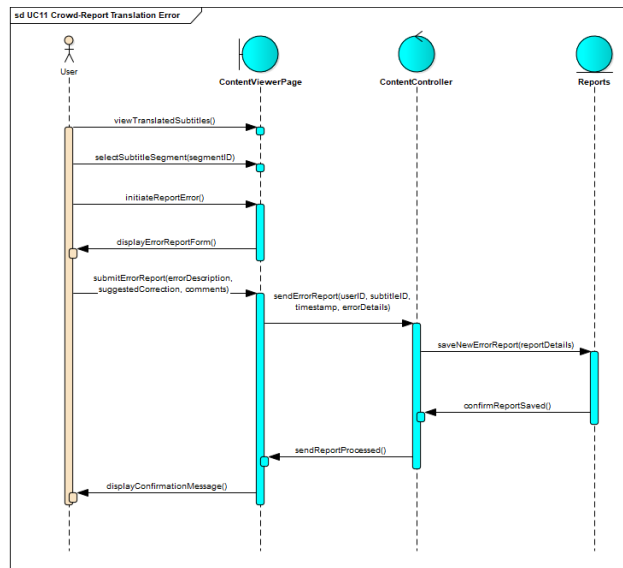


Figure 3.21: Sequence Diagram for <Crowd Report Translation Error>

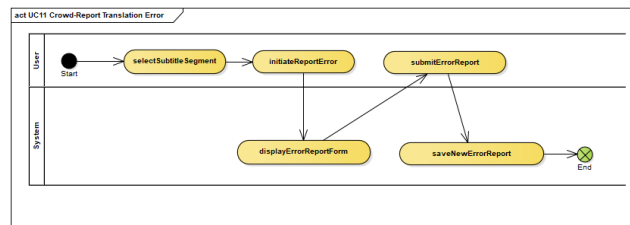


Figure 3.22: Activity Diagram for <Crowd Report Translation Error>

2.3.12 UC12: Use Case <Review Saved Vocabulary>

Use Case ID	UC12
Use Case Name	Review Saved Vocabulary
Description	This use case allows an authenticated user to access and review their personal list of saved vocabulary items, providing a centralized place for self-study and learning.
Actor(s)	User (Authenticated Language Learner or Educator)
Pre-condition(s)	1. The user is successfully logged into their LexiFlow account. 2. The user has previously saved at least one vocabulary item.
Normal Flow(s)- NF	1. The user navigates to the "My Vocabulary" or "Saved Words" section of their profile/dashboard. 2. The system retrieves the user's personal vocabulary list from the database. 3. The system displays the saved vocabulary items, typically including the word/phrase, its definition, pinyin, and the context from which it was saved.

	4. The user can browse, sort, or filter their vocabulary list.
Alternative Flow(s) - AF	AF1. Empty Vocabulary List: 3a. If the user's vocabulary list is empty: 3a1. The system displays a message indicating "No vocabulary saved yet" and may suggest how to save words (e.g., "Start watching videos and click on words to save!"). 3a2. The use case ends, providing guidance to the user.
Exception Flow(s) - EF	EF1. Database Retrieval Error: 3b. If the system encounters a technical error while retrieving the vocabulary list from the database: 3b1. The system logs the error. 3b2. The system displays a generic error message (e.g., "Could not load vocabulary. Please try again later."). 3b3. The use case terminates unsuccessfully.
Post-condition(s)	1. The user successfully views their saved vocabulary list. 2. The user can interact with the list (e.g., initiate flashcard generation, export).

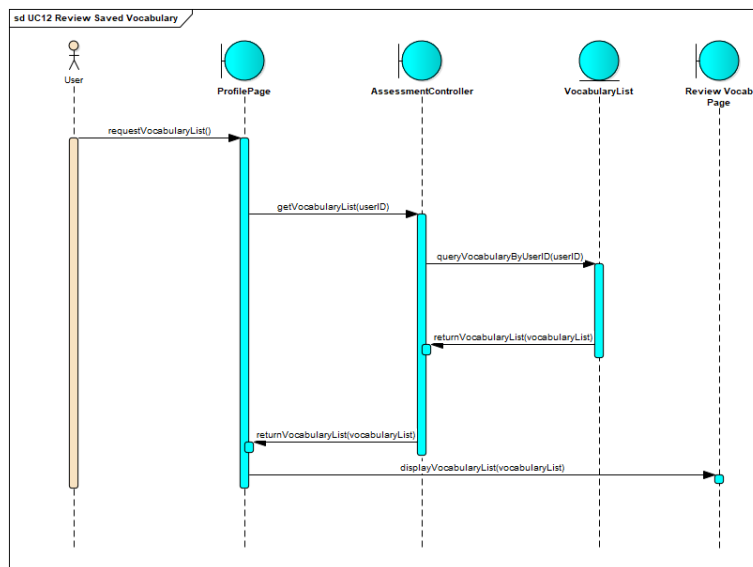


Figure 3.23: Sequence Diagram for <Review Saved Vocabulary>

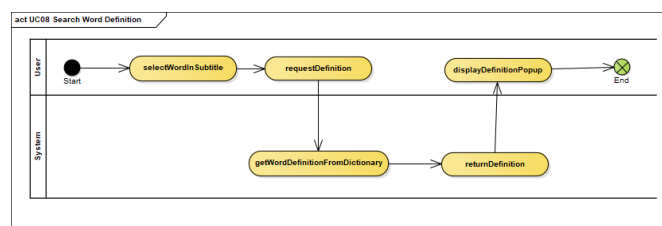


Figure 3.24: Activity Diagram for <Search Word Definition>

2.3.13 UC13: Use Case <Export Vocabulary List>

Use Case ID	UC13
Use Case Name	Export Vocabulary List
Description	This use case allows an authenticated user to export their personal saved vocabulary list into a common, portable file format (e.g., CSV, TXT), enabling external use or backup.
Actor(s)	User (Authenticated Language Learner or Educator)
Pre-condition(s)	1. The user is successfully logged into their LexiFlow account. 2. The user has at least one vocabulary item saved in their personal list.
Normal Flow(s)- NF	1. The user navigates to their "My Vocabulary" section. 2. The user selects an "Export Vocabulary" or similar option. 3. The system may prompt the user to choose an export format (e.g., CSV, TXT) and/or specify fields to include. 4. The user confirms the export options. 5. The system compiles the user's saved vocabulary data into the chosen file format. 6. The system initiates a file download to the user's local device. 7. The system confirms the successful export.
Alternative Flow(s) - AF	AF1. Empty Vocabulary List: 2a. If the user attempts to export an empty vocabulary list: 2a1. The system displays a message indicating "Your vocabulary list is empty. Nothing to export." 2a2. The use case ends.
Exception Flow(s) - EF	EF1. File Generation Error: 5a. If the system encounters a technical error while generating the export file: 5a1. The system logs the error. 5a2. The system displays an error message (e.g., "Could not export vocabulary. Please try again."). 5a3. The use case terminates unsuccessfully. EF2. Download Failure: 6a. If the download process fails due to network issues or browser restrictions: 6a1. The system displays a download error message. 6a2. The user is prompted to retry the download. 6a3. The use case terminates unsuccessfully.
Post-condition(s)	1. The user's saved vocabulary list is successfully exported and downloaded to their local device in the chosen file format.

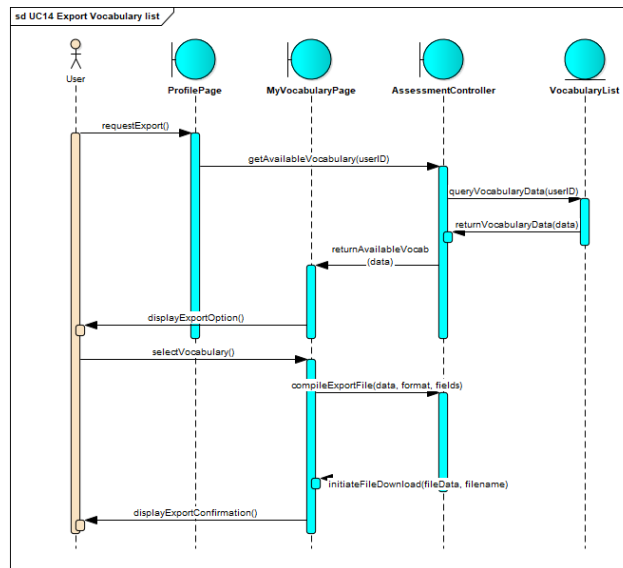


Figure 3.27: Sequence Diagram for <Export Vocabulary List>

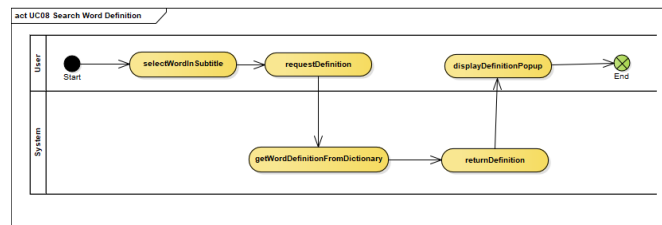


Figure 3.28: Activity Diagram for <Search Word Definition>

2.3.14 UC14: Use Case <Generate Quizzes>

Use Case ID	UC14
Use Case Name	Generate Quizzes
Description	This use case allows an authenticated user to generate customizable quizzes based on their saved vocabulary or specific content they have processed, facilitating self-assessment and reinforcement of learning.
Actor(s)	User (Authenticated Language Learner or Educator)
Pre-condition(s)	1. The user is successfully logged into their LexiFlow account. 2. The user has sufficient saved vocabulary or processed content available to generate a meaningful quiz.
Normal Flow(s)- NF	1. The user navigates to the "Quizzes" or "Learning Activities" section. 2. The user selects criteria for the quiz (e.g., source content, specific vocabulary tags, number of questions, question types like multiple-choice, matching, fill-in-the-blank). 3. The user initiates the "Generate Quiz" action. 4. The system compiles questions based on the selected criteria from saved vocabulary or processed content.

	5. The system presents the generated quiz to the user for completion. 6. The user completes the quiz and submits their answers. 7. The system scores the quiz and provides immediate feedback on correct/incorrect answers. 8. The system may store quiz results for tracking progress.
Alternative Flow(s) - AF	AF1. Insufficient Content/Vocabulary: 4a. If the selected criteria result in insufficient content or vocabulary to generate a quiz: 4a1. The system displays a message indicating "Not enough content/vocabulary for a quiz" and suggests refining criteria or saving more words. 4a2. The use case ends.
Exception Flow(s) - EF	EF1. Quiz Generation Failure: 4b. If the system encounters a technical error during the quiz generation process: 4b1. The system logs the error. 4b2. The system displays a generic error message (e.g., "Could not generate quiz. Please try again."). 4b3. The use case terminates unsuccessfully.
Post-condition(s)	1. A customizable quiz is successfully generated and presented to the user. 2. The user completes the quiz and receives immediate feedback on their performance. 3. (Optional) Quiz results are recorded for progress tracking.

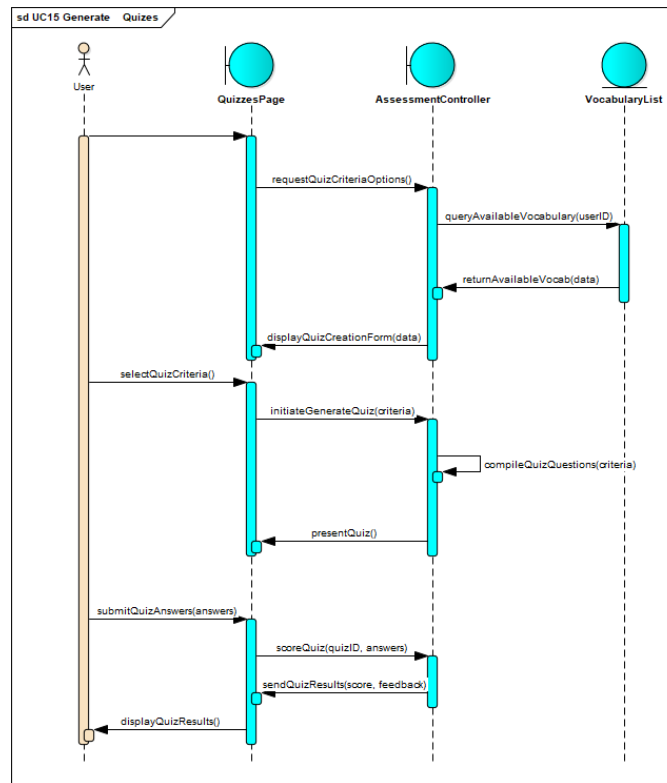


Figure 3.29: Sequence Diagram for <Search Word Definition>

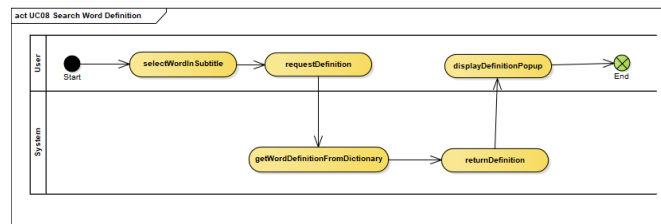


Figure 3.30: Activity Diagram for <Search Word Definition>

2.4 Performance and Other Requirements

ID	Description	Related Requirement
NFR-001	The system shall process submitted audio/video content (voice activity detection, transcription, translation, and pinyin annotation) within 10 minutes of processing time per hour of source-content duration. Verified against 16 production processing runs; see Chapter 5 (Non-Functional Requirement Verification) of the accompanying thesis.	UC04 Process YouTube URL, UC05 Upload Local Media, UC06 Generate Subtitles

2.5 Design Constraints

ID	Constraint Description
DC-01	The system shall utilize cloud-based services (e.g., Google Cloud Platform, AWS, Azure) for scalability and resource management.
DC-02	The front-end user interface shall be developed using web technologies (e.g., HTML, CSS, JavaScript frameworks like React or Vue.js).
DC-03	The back-end shall be implemented using Python for its rich ecosystem of AI/ML libraries.
DC-04	The database system shall be a NoSQL database (e.g., MongoDB, Firestore) to accommodate semi-structured subtitle data and enable flexible querying.
DC-05	The system's design must prioritize modularity to allow for easy integration of future language pairs and AI model updates.